

1 Prognostic Nutritional Index and All-Cause Mortality in
2 Gastrointestinal Cancer: A Dual-Cohort NHANES
3 Analysis With Time-Dependent, Competing Risk, and
4 Joint Inflammatory Assessment

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6 **Abstract**

7 **Background & Aims:** Nutritional status is a modifiable prognostic factor
8 in cancer patients, but population-level validation of composite nutritional in-
9 dices in gastrointestinal (GI) cancer remains limited. We aimed to validate the
10 Prognostic Nutritional Index (PNI), Controlling Nutritional Status (CONUT),
11 and Geriatric Nutritional Risk Index (GNRI) as prognostic factors for all-cause
12 mortality in GI cancer patients using nationally representative data.

13 **Methods:** We identified 313 GI cancer patients from NHANES III (1988–
14 1994, n=92) and Continuous NHANES (2005–2016, n=221) with mortality
15 follow-up through December 2019 (up to 31 years). PNI, CONUT, and GNRI
16 were derived from baseline serum albumin, lymphocyte count, total cholesterol,
17 and body mass index. Cox proportional hazards, time-dependent Cox, landmark
18 analysis, competing risks, restricted cubic splines, restricted mean survival time
19 (RMST), and population attributable fraction (PAF) analyses were performed.
20 E-value sensitivity analysis quantified robustness to unmeasured confounding.

21 **Results:** Higher PNI was associated with reduced all-cause mortality (ad-
22 justed HR 0.78, 95% CI 0.63–0.96, p=0.020). GNRI showed the strongest associ-
23 ation (HR 0.67, 95% CI 0.57–0.80, p<0.001). The PNI effect was strongly time-

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24 dependent, maximal in the first 2 years (HR 0.34, 95% CI 0.21–0.54, $p < 0.001$)
 25 and attenuating thereafter (interaction $p < 0.001$). Optimal PNI threshold was
 26 48.5 (cross-validated mean 48.3); patients below this had median survival 6.6
 27 vs 12.8 years ($p < 0.0001$) with RMST difference 1.97 years (95% CI 1.19–2.76,
 28 $p < 0.001$). In competing risks, PNI primarily protected against non-cancer death
 29 (HR 0.63, $p = 0.001$) rather than GI cancer-specific death (HR 1.06, $p = 0.700$).
 30 PNI decomposition showed albumin alone outperformed the composite index
 31 ($\Delta C\text{-stat} +0.032$ vs $+0.022$), while lymphocyte count contributed no indepen-
 32 dent prognostic information (HR 1.06, $p = 0.540$). Dietary quality (HEI-2015)
 33 did not predict survival (HR 0.97, $p = 0.850$). Inflammatory markers (NLR,
 34 SII) showed comparable prognostic performance to PNI. Results were consis-
 35 tent across NHANES generations (cohort interaction $p = 0.480$) and robust to
 36 adjustment for education, income, smoking, and comorbidity. Fine-Gray sub-
 37 distribution hazard models confirmed the competing risks patterns. CONUT
 38 and GNRI survived Bonferroni correction for multiple testing ($\alpha = 0.0083$),
 39 while PNI (adjusted $p = 0.020$) did not.

40 **Conclusions:** CONUT and GNRI are robust, independent prognostic fac-
 41 tors for all-cause mortality in GI cancer patients, validated across two nationally
 42 representative US cohorts spanning 30 years. PNI showed a nominally signifi-
 43 cant association that did not survive Bonferroni correction and requires inde-
 44 pendent replication. The protective effects are time-dependent, operate primar-
 45 ily through non-cancer mortality pathways, and are driven by serum albumin.
 46 These simple, inexpensive indices may guide early nutritional intervention in GI
 47 cancer patients.

48 *Keywords:* Prognostic Nutritional Index, Gastrointestinal Cancer, NHANES,
 49 Survival Analysis, Competing Risks, Time-Dependent Effect

50 1. Introduction

51 Cancer cachexia and malnutrition represent critical challenges in gastroin-
52 testinal oncology. GI cancer patients are uniquely vulnerable due to tumor-
53 induced catabolism, malabsorption, and mechanical obstruction, with malnutri-
54 tion prevalence ranging from 30% to 80% depending on tumor site and stage [1,
55 2, 3]. Malnourished patients consistently show poorer treatment tolerance, in-
56 creased postoperative complications, and reduced survival [4, 5].

57 Several composite nutritional indices have been developed to integrate multi-
58 ple biomarkers into a single prognostic score. The Prognostic Nutritional Index
59 (PNI), originally proposed by Onodera et al. in 1984, combines serum albu-
60 min and peripheral lymphocyte count [6]. The Controlling Nutritional Sta-
61 tus (CONUT) score incorporates albumin, lymphocyte count, and total cholest-
62 erol [7]. The Geriatric Nutritional Risk Index (GNRI) adjusts albumin for body
63 mass index [8]. Each index has been validated primarily in single-center surgi-
64 cal series, predominantly in East Asian populations with evidence concentrated
65 in specific GI cancer subtypes such as gastric, colorectal, and hepatocellular
66 carcinoma [9, 10, 11].

67 Despite their clinical utility, important gaps remain. First, population-level
68 validation across different geographic contexts and survey eras is lacking. Most
69 prior studies originate from single-institution cohorts with sample sizes rarely
70 exceeding 1,000 patients [12, 13, 14]. Second, the temporal dynamics of nutri-
71 tional risk—whether the prognostic effect is constant or concentrated in specific
72 post-diagnosis windows—have not been systematically examined using time-
73 dependent analyses. Third, the conceptual distinction between biochemical
74 nutritional status (serum proteins, lymphocyte count) and dietary nutritional
75 quality (nutrient intake patterns) remains poorly characterized in cancer prog-
76 nosis.

77 To address these gaps, we undertook a comprehensive analysis of nutritional
78 indices in GI cancer patients using the National Health and Nutrition Exami-
79 nation Survey (NHANES), a nationally representative population-based survey
80 with mortality linkage. We leveraged two independent NHANES generations—
81 NHANES III (1988–1994) and Continuous NHANES (2005–2016)—providing
82 up to 31 years of mortality follow-up and enabling cross-validation across differ-
83 ent survey eras. Our objectives were to: (1) validate PNI, CONUT, and GNRI
84 as prognostic factors for all-cause mortality; (2) characterize time-dependent ef-
85 fects through landmark, time-dependent Cox, and RMST analyses; (3) compare
86 prognostic performance across competing causes of death; and (4) compare bio-
87 chemical nutritional indices against a dietary quality measure (HEI-2015) and
88 inflammatory markers (NLR, SII).

89 **2. Materials and Methods**

90 *2.1. Study Population*

91 We used data from two independent NHANES cycles: NHANES III (1988–
92 1994) and Continuous NHANES (2005–2016). NHANES is a nationally repre-
93 sentative cross-sectional survey of the civilian non-institutionalized US popula-
94 tion conducted by the National Center for Health Statistics (NCHS). Both sur-
95 veys collected demographic, dietary, laboratory, and examination data through
96 standardized protocols. The analytic sample included participants aged ≥ 18
97 years with available nutrition biomarker data. A total of 44,066 participants
98 met inclusion criteria, of whom 313 had a GI cancer diagnosis.

99 The NHANES protocol was approved by the NCHS Ethics Review Board
100 (Protocol #2005-06 for 2005–2016 cycles; Protocol #98-12 and Institutional Re-
101 view Board approval for NHANES III). Written informed consent was obtained
102 from all participants. As a secondary analysis of de-identified public-use data,

103 this study was exempt from additional institutional review board approval.

104 2.2. *GI Cancer Ascertainment*

105 GI cancer was identified from self-reported cancer history. In Continuous
106 NHANES (2005–2016), cancer diagnoses were identified through the Medical
107 Conditions Questionnaire (MCQ) using cancer site codes. In NHANES III,
108 cancer site was identified from variable HAC3OS, with GI sites defined as codes
109 for colon, stomach, esophagus, pancreas, liver, and gallbladder. GI cancer was
110 defined as cancer of the esophagus, stomach, colon, rectum, pancreas, liver, or
111 gallbladder.

112 2.3. *Nutritional Indices*

113 **PNI** was calculated according to Onodera’s formula: $\text{PNI} = 10 \times \text{serum}$
114 $\text{albumin (g/dL)} + 0.005 \times \text{lymphocyte count } (\mu\text{L}^{-1})$ [6].

115 **CONUT** was derived as the sum of three component scores: albumin (≥ 3.5
116 $\text{g/dL} = 0$, $3.0\text{--}3.49 = 2$, $2.5\text{--}2.99 = 4$, $< 2.5 = 6$), lymphocyte count (≥ 1600
117 $\mu\text{L}^{-1} = 0$, $1200\text{--}1599 = 1$, $800\text{--}1199 = 2$, $< 800 = 3$), and total cholesterol (≥ 180
118 $\text{mg/dL} = 0$, $140\text{--}179 = 1$, $100\text{--}139 = 2$, $< 100 = 3$) [7].

119 **GNRI** was calculated as: $\text{GNRI} = 14.89 \times \text{albumin (g/dL)} + 41.7 \times (\text{BMI}$
120 $/ 22)$ [8].

121 Serum albumin was measured using the bromcresol purple method in Con-
122 tinuous NHANES and bromcresol green in NHANES III, with g/L values con-
123 verted to g/dL for consistency. Complete blood count provided lymphocyte
124 counts. Total cholesterol was measured enzymatically.

125 **HEI-2015** was calculated from 24-hour dietary recall data using the Food
126 Patterns Equivalents Database, measuring adherence to the 2015–2020 Dietary
127 Guidelines for Americans on a 0–100 scale.

128 2.4. Mortality Linkage

129 Mortality status was ascertained through linkage to the NCHS National
130 Death Index through December 31, 2019. Follow-up time was calculated from
131 the date of survey participation to death or censoring. The underlying cause
132 of death was classified using the UCOD_LEADING variable, with malignant
133 neoplasms (code 002) indicating cancer death.

134 2.5. Covariates

135 Demographic covariates included age (continuous, years), sex (male/female),
136 and race/ethnicity (Non-Hispanic White, Non-Hispanic Black, Hispanic, Other),
137 selected a priori based on known associations with both nutritional status and
138 mortality. Extended models additionally adjusted for education (high school
139 or above), income-to-poverty ratio, and physical activity (any moderate-to-
140 vigorous activity). Cohort (NHANES III vs. Continuous NHANES) was in-
141 cluded as a fixed effect to account for differences in albumin assay methodology
142 between survey eras.

143 2.6. Statistical Analysis

144 All analyses were conducted using R 4.6.0 with the `survey`, `survival`,
145 `EValue`, `survRM2`, and `broom` packages.

146 **Survival analysis:** Cox proportional hazards models estimated hazard ra-
147 tios (HR) per 1-standard deviation (SD) increase in each nutritional index.
148 Both crude and adjusted models (age, sex, race/ethnicity) were fitted. The pro-
149 portional hazards assumption was tested using Schoenfeld residuals. For PNI,
150 which violated this assumption ($p=0.017$), time-dependent Cox models were
151 fitted with time splitting at 2 and 5 years.

152 **Time-dependent effects:** Landmark analyses at 1, 3, and 5 years exam-
153 ined conditional survival. Restricted mean survival time (RMST) at 10 years

154 was compared between high and low PNI groups using the threshold identified
155 by maximally selected log-rank statistics [15, 16]. Restricted cubic splines (3
156 knots) examined nonlinear dose-response relationships.

157 **Competing risks:** Cause-specific Cox models were fitted for GI cancer
158 death vs. non-cancer death [17, 18]. Subdistribution hazard models (Fine-Gray)
159 were additionally fitted to address the competing risk structure from an absolute
160 risk perspective [19].

161 **Component analysis:** PNI was decomposed into albumin alone and lym-
162 phocyte alone, comparing their prognostic performance to the composite index
163 using AIC, change in C-statistic with 2000 bootstrap replications, and likelihood
164 ratio tests.

165 **Cohort-stratified and interaction analyses:** Cohort-stratified models
166 (NHANES III vs. Continuous NHANES) examined effect consistency. Formal
167 cohort-by-index interaction terms tested heterogeneity. Cohort was also in-
168 cluded as a fixed effect to address albumin assay differences (bromocresol green
169 in NHANES III vs. bromocresol purple in Continuous NHANES).

170 **Extended covariate models:** Sensitivity analyses additionally adjusted
171 for education (high school or above), income-to-poverty ratio, smoking status
172 (current vs. non-current), and a simplified Charlson comorbidity score (myocar-
173 dial infarction, congestive heart failure, stroke, chronic obstructive pulmonary
174 disease, liver disease, and diabetes) to assess robustness to socioeconomic, be-
175 havioral, and comorbidity confounding. Multiple testing was addressed using
176 Bonferroni correction for the primary family of six tests (three indices $\times$ two
177 adjustment levels).

178 **Reverse causation sensitivity:** Analyses excluding deaths within the first
179 2 years of follow-up addressed potential reverse causation from undiagnosed
180 advanced disease at baseline.

181 **Joint exposure analysis:** PNI and NLR were dichotomized at medians
182 into four groups, with additive interaction quantified using Relative Excess Risk
183 due to Interaction (RERI) with bootstrap 95% CI.

184 **Population attributable fraction (PAF):** $PAF = p \times (HR - 1) / HR$
185 was estimated for $PNI < 48.5$ with bootstrap CI [20].

186 **E-value sensitivity analysis:** E-values quantified the minimum associa-
187 tion strength an unmeasured confounder would need to fully explain away the
188 observed results [21].

189 **Bootstrap C-statistic:** Incremental discrimination (ΔC -stat) over base-
190 line covariates was estimated with 500 bootstrap replications.

191 **Inflammatory markers:** NLR and SII were compared with PNI in the
192 2005–2016 subset with available neutrophil and platelet counts.

193 3. Results

194 3.1. Baseline Characteristics

195 Of 44,066 eligible participants aged ≥ 18 , 313 (0.7%) had a GI cancer diagno-
196 sis, comprising 92 from NHANES III and 221 from Continuous NHANES. Col-
197 orectal cancer was the most common ($n=241$, 77.0%), followed by liver ($n=23$,
198 7.3%), stomach ($n=19$, 6.1%), esophageal ($n=15$, 4.8%), pancreatic ($n=8$, 2.6%),
199 and gallbladder ($n=6$, 1.9%). The remaining cases were rectal ($n=8$). GI can-
200 cer patients were substantially older than the general population (67.5 vs 45.5
201 years) and had lower PNI (50.8 vs 53.2), indicating worse nutritional status.
202 During a median follow-up of 8.8 years (up to 31 years), 169 deaths (54.0%)
203 occurred among GI cancer patients.

204 Table 1 summarizes baseline characteristics by GI tumor status, and Table 2
205 shows characteristics by PNI tertile among GI cancer patients.

Table 1: Baseline characteristics by GI tumor status (NHANES III + Continuous NHANES 2005–2016).

Variable	GI Tumor (n=313)	Non-Cancer (n=40,480)
Age, years	67.5 $\pm$ 13.8	45.5 $\pm$ 18.4
Female (%)	64 (20.4)	7,172 (17.7)
Non-Hispanic White (%)	160 (51.1)	14,887 (36.8)
Education $\geq$ high school (%)	187 (59.7)	25,808 (63.8)
BMI, kg/m ²	28.5 $\pm$ 6.1	28.3 $\pm$ 6.6
Albumin, g/dL	4.1 $\pm$ 0.4	4.2 $\pm$ 0.4
Lymphocyte, μL^{-1}	1,972 $\pm$ 990	2,209 $\pm$ 719
Total cholesterol, mg/dL	195 $\pm$ 44	196 $\pm$ 43
PNI	50.8 $\pm$ 6.5	53.2 $\pm$ 5.2
CONUT	1.1 $\pm$ 1.4	0.7 $\pm$ 1.0
GNRI	115.0 $\pm$ 12.2	116.4 $\pm$ 12.4
Deaths (%)	169 (54.0)	7,659 (18.9)

Table 2: Baseline characteristics by PNI tertile among GI cancer patients.

Variable	Low PNI (T1)	Mid PNI (T2)	High PNI (T3)
N	105	104	104
Age, years	72.6 $\pm$ 9.4	66.4 $\pm$ 15.5	63.6 $\pm$ 13.8
Female (%)	15 (14.3)	26 (25.0)	23 (22.1)
PNI	44.8 $\pm$ 3.5	50.5 $\pm$ 1.4	57.5 $\pm$ 4.1
CONUT	2.1 $\pm$ 1.7	0.9 $\pm$ 0.9	0.3 $\pm$ 0.5
GNRI	110.2 $\pm$ 12.6	115.5 $\pm$ 10.4	119.5 $\pm$ 11.7
Albumin, g/dL	3.8 $\pm$ 0.4	4.1 $\pm$ 0.3	4.4 $\pm$ 0.3
Lymphocyte, μL^{-1}	1,376 $\pm$ 389	1,820 $\pm$ 516	2,739 $\pm$ 1,049
Total cholesterol, mg/dL	184 $\pm$ 45	196 $\pm$ 41	205 $\pm$ 43
BMI, kg/m ²	28.3 $\pm$ 6.1	28.5 $\pm$ 5.6	28.7 $\pm$ 6.6
Deaths (%)	74 (70.5)	48 (46.2)	47 (45.2)
Median survival	5.5 yr	7.8 yr	13.9 yr

Patients in the lowest PNI tertile had substantially higher mortality (70.5%) compared to the middle (46.2%) and highest (45.2%) tertiles, with median survival of 5.5, 7.8, and 13.9 years, respectively (Figure 1A).

3.2. All-Cause Mortality

All three nutritional indices significantly predicted all-cause mortality after adjustment for age, sex, and race/ethnicity (Table 3). GNRI showed the strongest association (HR 0.67 per 1-SD, 95% CI 0.57–0.80, $p < 0.001$), followed by CONUT (HR 0.71, 95% CI 0.61–0.83, $p < 0.001$) and PNI (HR 0.78, 95% CI 0.63–0.96, $p = 0.020$) (Figure 2A).

Table 3: Cox proportional hazards—nutritional indices and all-cause mortality.

Index	Adjustment	HR (95% CI) per 1-SD	p-value
PNI	Crude	0.63 (0.51–0.78)	< 0.001
PNI	Adjusted*	0.78 (0.63–0.96)	0.020
CONUT	Crude	0.63 (0.55–0.73)	< 0.001
CONUT	Adjusted*	0.71 (0.61–0.83)	< 0.001
GNRI	Crude	0.67 (0.56–0.79)	< 0.001
GNRI	Adjusted*	0.67 (0.57–0.80)	< 0.001

*Adjusted for age, sex, race/ethnicity.

3.3. Time-Dependent Effects

The proportional hazards assumption was violated for PNI (Schoenfeld residuals $p = 0.017$) but not for GNRI ($p = 0.087$) or CONUT ($p = 0.430$). Time-dependent Cox models revealed marked attenuation of the PNI effect: the protective association was strongest in the first 2 years (HR 0.34, 95% CI 0.21–0.54, $p < 0.001$) and non-significant thereafter (2–5 year: HR 1.32, $p = 0.490$; 5+ year: HR 0.81, $p = 0.450$; interaction $p < 0.001$). GNRI showed a similar temporal pattern (0–2 year HR 0.53, $p = 0.034$), while CONUT demonstrated more stable effects across periods (Figure 3A).

224 In landmark analysis, PNI showed attenuated and non-significant effects at
 225 1, 3, and 5 years, while CONUT and GNRI maintained significant associations
 226 at all landmark times (Table 4, Figure 3B).

Table 4: Landmark analysis—adjusted HR per 1-SD.

Index	1-year	3-year	5-year
PNI	0.91 (0.73–1.13, p=0.380)	0.85 (0.66–1.10, p=0.210)	0.81 (0.60–1.09, p=0.170)
CONUT	0.75 (0.63–0.89, p<0.001)	0.71 (0.58–0.88, p=0.002)	0.74 (0.57–0.96, p=0.021)
GNRI	0.70 (0.59–0.84, p<0.001)	0.70 (0.57–0.86, p<0.001)	0.70 (0.56–0.89, p=0.003)

227 3.4. Optimal PNI Threshold and Dose-Response

228 Maximally selected log-rank statistics identified **PNI < 48.5** as the optimal
 229 prognostic threshold (p<0.0001). Patients below this threshold had markedly
 230 shorter median survival (6.6 vs 12.8 years) (Figure 1B). RMST analysis at 10
 231 years confirmed a clinically meaningful difference of 1.97 years between groups
 232 (95% CI 1.19–2.76, p<0.001), with PAF indicating 13.4% of deaths attributable
 233 to low PNI (95% CI 6.2%–20.1%).

234 Restricted cubic spline analysis confirmed a nonlinear relationship between
 235 PNI and mortality risk (nonlinearity p<0.0001), with sharp risk increase below
 236 approximately 45 and a plateau above 50 (Figure 4). In contrast, CONUT
 237 and GNRI showed linear dose-response relationships (nonlinearity p=0.110 and
 238 p=0.160, respectively).

239 3.5. Competing Risks Analysis

240 Cause-specific Cox models revealed divergent patterns (Figure 2B). PNI
 241 primarily protected against non-cancer death (HR 0.63, 95% CI 0.48–0.83,
 242 p=0.001) rather than GI cancer-specific death (HR 1.06, 95% CI 0.78–1.45,
 243 p=0.700). CONUT and GNRI showed protective effects against both cancer-
 244 specific and non-cancer death (Table 5).

Table 5: Competing risks—cause-specific HR per 1-SD.

Index	GI Cancer Death	Non-Cancer Death
PNI	1.06 (0.78–1.45, p=0.700)	0.63 (0.48–0.83, p=0.001)
CONUT	0.73 (0.56–0.94, p=0.016)	0.70 (0.58–0.85, p<0.001)
GNRI	0.69 (0.51–0.94, p=0.017)	0.66 (0.53–0.82, p<0.001)

For GI cancer death, Fine-Gray subdistribution hazard models showed no index reaching statistical significance (PNI sHR 1.16, 95% CI 0.77–1.75, p=0.490; CONUT sHR 0.82, 95% CI 0.59–1.14, p=0.240; GNRI sHR 0.79, 95% CI 0.58–1.07, p=0.120). For non-cancer death, GNRI remained significant (sHR 0.80, 95% CI 0.64–0.99, p=0.039) and PNI was borderline (sHR 0.81, 95% CI 0.66–1.00, p=0.050).

3.6. PNI vs Inflammatory Markers

In the subset with neutrophil and platelet counts (n=258, 125 events), NLR (HR 1.43 per 1-SD, 95% CI 1.19–1.72, p<0.001) and SII (HR 1.30, 95% CI 1.10–1.54, p=0.002) significantly predicted mortality. These findings are consistent with prior meta-analyses demonstrating NLR as a robust prognostic marker across solid tumors [22]. C-statistic improvement was comparable across indices (+0.017 for PNI, +0.017 for logNLR). In mutually adjusted models including both PNI and logNLR, neither remained significant (PNI HR 0.86, p=0.290; logNLR HR 1.22, p=0.090), indicating shared predictive information.

3.7. Joint PNI $\times$ NLR Analysis

The “Low PNI + High NLR” group had significantly higher mortality (HR 1.67, 95% CI 1.06–2.63, p=0.028) compared to the “High PNI + Low NLR” reference group. RERI was -0.97 (95% CI -2.61 to -0.10), suggesting overlapping rather than synergistic pathways.

265 3.8. CRP Interaction and CALLY Index

266 Among 250 patients with CRP measurements, CRP was non-significantly
267 associated with mortality (HR 1.18 per log-SD, $p=0.190$). No PNI $\times$ CRP
268 interaction was observed ($p=0.780$). The CALLY index (albumin $\times$ lymphocyte
269 / CRP) did not outperform PNI (CALLY HR 0.85, $p=0.200$; C-stat 0.659 vs
270 0.667 for PNI), despite prior studies reporting its prognostic value in colorectal
271 cancer populations [23].

272 3.9. Dietary Quality Comparison

273 Among 99 GI cancer patients with dietary data, HEI-2015 did not predict
274 mortality (HR 0.97, $p=0.850$). The correlation between HEI-2015 and PNI
275 was weak ($r=0.16$), confirming these measures capture distinct dimensions of
276 nutritional status.

277 3.10. Subgroup and Sensitivity Analyses

278 PNI showed directionally consistent protective effects across all examined
279 subgroups, with no significant effect modification by age (interaction $p=0.730$),
280 sex ($p=0.340$), or BMI ($p=0.680$) (Figure 6). Bootstrap C-statistic (500 repli-
281 cations) showed GNRI had the largest Δ C-stat (+0.034), followed by CONUT
282 (+0.023) and PNI (+0.022). Survey-weighted Cox (2005–2016 subset) atten-
283 uated PNI (HR 0.87, $p=0.430$) while CONUT remained significant (HR 0.76,
284 $p=0.003$). E-values ranged from 1.66 (PNI) to 1.97 (GNRI), indicating moder-
285 ate robustness to unmeasured confounding.

286 3.11. Cohort Consistency and Extended Sensitivity Analyses

287 Cohort-stratified analyses revealed consistent effects across NHANES gen-
288 erations. PNI showed comparable associations in NHANES III (HR 0.69, 95%
289 CI 0.51–0.93, $p=0.015$) and Continuous NHANES (HR 0.75, 95% CI 0.56–1.00,

p=0.047), with no significant cohort-by-PNI interaction (p=0.480). Similar consistency was observed for CONUT (interaction p=0.940) and GNRI (interaction p=0.950), supporting the appropriateness of pooling these cohorts. Inclusion of cohort as a fixed effect did not materially alter the results (e.g., PNI HR 0.73, 95% CI 0.59–0.89).

Adjustment for additional socioeconomic covariates (education, income-to-poverty ratio) modestly strengthened the PNI association (HR 0.74, 95% CI 0.55–1.00, p=0.047) and did not attenuate CONUT or GNRI. Further adjustment for smoking status and Charlson comorbidity score in the Continuous NHANES subset with available data (N=133, 61 events) did not attenuate the associations: PNI HR 0.68 (95% CI 0.48–0.97, p=0.035), CONUT HR 0.63 (95% CI 0.50–0.80, p<0.001), and GNRI HR 0.72 (95% CI 0.55–0.93, p=0.013). The stability of the effect estimates across progressive adjustment for socioeconomic, behavioral, and comorbidity confounders strengthens the evidence for independent prognostic value, though residual confounding by unmeasured factors remains possible.

In sensitivity analyses excluding deaths within 2 years (n=284, 140 events), PNI was no longer associated with mortality (HR 0.98, 95% CI 0.78–1.23, p=0.830), confirming that its prognostic effect is concentrated in the early post-assessment period. In contrast, CONUT and GNRI maintained significant associations (HR 0.77, p=0.006 and HR 0.70, p<0.001, respectively), suggesting more stable long-term prognostic value.

3.12. Multiple Testing Consideration

In the primary analysis family of six tests (three indices $\times$ crude/adjusted models), the Bonferroni-corrected significance threshold was $\alpha = 0.0083$. PNI adjusted (p=0.020) did not survive this correction, while CONUT and GNRI adjusted (both p<0.001) and all crude models (all p<0.001) remained signifi-

317 cant. These results are therefore reported as primary analyses for CONUT and
318 GNRI, with PNI interpreted as a secondary finding requiring replication.

319 3.13. PNI Decomposition

320 Formal comparison of PNI against its individual components confirmed that
321 the prognostic value was driven entirely by serum albumin (Figure 5B). Albumin
322 alone achieved a lower AIC (1595.8 vs. 1607.8) and larger bootstrap C-statistic
323 improvement (Δ C-stat +0.032, 95% CI 0.010–0.063) than the composite PNI
324 (Δ C-stat +0.022, 95% CI –0.001–0.051). Lymphocyte count alone provided
325 negligible discrimination (Δ C-stat +0.002, 95% CI –0.007–0.014). A likelihood
326 ratio test confirmed that adding lymphocyte count to an albumin-only model
327 did not improve fit ($p=0.460$), indicating that lymphocyte count contributes no
328 independent prognostic information beyond albumin in this population.

329 4. Discussion

330 4.1. Summary of Principal Findings

331 In this population-based dual-cohort analysis spanning 30 years of NHANES
332 data, we demonstrate that CONUT and GNRI consistently predict all-cause
333 mortality in GI cancer patients after Bonferroni correction, while PNI showed a
334 nominally significant but Bonferroni-non-significant association ($p=0.020$) that
335 requires replication. GNRI showed the strongest association (HR 0.67 per 1-SD,
336 95% CI 0.57–0.80), followed by CONUT (HR 0.71, 95% CI 0.61–0.83) and PNI
337 (HR 0.78, 95% CI 0.63–0.96). The PNI effect was strongly time-dependent, max-
338 imal in the first 2 years (HR 0.34), and operated primarily through non-cancer
339 mortality rather than GI cancer-specific pathways. Critically, the prognostic
340 value was driven by serum albumin alone, with lymphocyte count contribut-
341 ing no independent information. RMST analysis quantified a 1.97-year survival
342 difference between high and low PNI groups at 10 years.

343 4.2. Time-Dependent and Cause-Specific Effects

344 CONUT and GNRI showed stable, time-independent associations with all-
345 cause mortality across the full follow-up period. In landmark analyses, both
346 remained significant at 1, 3, and 5 years, and the proportional hazards assumption
347 was not violated for CONUT ($p=0.430$) or GNRI ($p=0.087$).

348 The temporal variation of PNI’s prognostic effect—maximal in the first 2
349 years (HR 0.34) followed by marked attenuation to null after excluding early
350 deaths—stands in contrast to the stability of CONUT and GNRI. Pre-existing
351 nutritional status most strongly influences short-term survival, likely through its
352 impact on treatment tolerance, perioperative recovery, and therapy completion.
353 This pattern aligns with prior observations in a Korean gastric cancer cohort [24]
354 and suggests that the window for effective nutritional intervention is early in the
355 cancer trajectory. The complete attenuation after 2 years (HR 0.98 in the >2-
356 year sensitivity analysis) also suggests that a single baseline PNI measurement
357 cannot capture the dynamic nature of nutritional status over extended follow-
358 up.

359 The competing risks analysis provides mechanistic insight: PNI primarily
360 protects against non-cancer death (HR 0.63, $p=0.001$) rather than GI cancer-
361 specific death (HR 1.06, $p=0.700$), suggesting PNI captures physiological reserve
362 and resilience to treatment-related complications rather than directly influencing
363 tumor biology. This distinction is critical for clinical nutrition: it aligns with
364 ESPEN guidelines emphasizing that nutritional therapy aims to maintain capacity
365 for anti-cancer treatment and reduce non-cancer complications, rather than
366 to modify cancer biology per se [1, 2]. Patients with low PNI may benefit from
367 intensified supportive care and closer monitoring for non-cancer complications.
368 CONUT and GNRI showed protective effects against both cancer-specific and
369 non-cancer death, possibly reflecting the additional prognostic information con-

tributed by total cholesterol and BMI, which may capture aspects of metabolic health relevant to cancer progression.

4.3. Nutritional vs Inflammatory Pathways

PNI decomposition revealed that serum albumin alone (HR 0.69, 95% CI 0.58–0.82) outperformed the composite PNI (HR 0.78) across multiple metrics: lower AIC (1595.8 vs. 1607.8), larger C-statistic improvement, and significantly better model fit (likelihood ratio test $p < 0.001$), while lymphocyte count showed no independent prognostic value (Δ C-stat +0.002, 95% CI –0.007–0.014). This finding calls into question whether the lymphocyte component of PNI adds meaningful prognostic information beyond albumin alone in this population. The biological rationale for PNI’s association with mortality appears to be mediated primarily by albumin’s dual role as a nutritional marker and negative acute-phase reactant reflecting systemic inflammation, rather than by the immunonutritional synergy PNI was originally designed to capture.

In the context of current consensus frameworks, PNI and CONUT may be positioned as screening tools that identify patients warranting formal GLIM (Global Leadership Initiative on Malnutrition) assessment. While PNI components (albumin, lymphocyte count, total cholesterol) correlate with GLIM etiologic criteria (disease burden/inflammation), they do not substitute for the full GLIM phenotypic assessment including weight loss, low BMI, and reduced muscle mass.

Inflammatory markers NLR and SII showed comparable prognostic performance to PNI (+0.017 C-statistic improvement for both). The mutually adjusted model, in which neither PNI nor NLR remained significant, indicates substantial overlap between nutritional and inflammatory signaling pathways. The joint exposure analysis (Low PNI + High NLR, HR 1.67, 95% CI 1.06–2.63; RERI –0.97) suggests that combined assessment identifies the highest-risk pa-

397 tients, though the negative RERI indicates overlapping rather than synergistic
398 pathways.

399 Notably, the absence of PNI $\times$ CRP interaction ($p=0.780$) and the non-
400 significant CALLY index finding (HR 0.85, $p=0.200$) indicate that PNI's prog-
401 nostic value extends beyond what can be explained by systemic inflammation
402 measured through single biomarkers. The null dietary finding (HEI-2015 HR
403 0.97, $p=0.850$), though limited by reliance on a single 24-hour dietary recall
404 rather than usual intake estimates, reinforces the conceptual distinction be-
405 tween biochemical nutritional status—reflecting metabolic reserve and systemic
406 inflammation—and dietary intake patterns, which may be less relevant in the
407 post-diagnosis period when metabolic alterations dominate.

408 4.4. *Comparison with Previous Literature*

409 Our findings extend prior single-center studies validating PNI in GI can-
410 cer. A meta-analysis of 28 studies in gastric cancer reported that low PNI was
411 significantly associated with poorer overall survival (HR 1.38, 95% CI 1.23–
412 1.54) [9], consistent with our population-level results. More recently, analyses
413 using NHANES data have similarly demonstrated PNI's prognostic value for
414 cancer-specific and all-cause mortality in US populations [25], corroborating
415 our findings in an independent sample. A novel contribution of our study is
416 the demonstration that the PNI effect is consistent across two NHANES gen-
417 erations (cohort interaction $p=0.480$), supporting generalizability across survey
418 eras. Our study uniquely provides dual-cohort cross-validation across 30 years,
419 absolute risk estimates (RMST) that enhance clinical interpretability, and time-
420 dependent analyses demonstrating the temporal window of nutritional risk. The
421 head-to-head comparison of PNI, CONUT, and GNRI—rarely performed in a
422 single population—identifies GNRI as the most stable and strongly associated
423 index, likely because BMI adjustment captures the paradoxical protective effect

of preserved body mass in cancer patients. Notably, the optimal PNI threshold of 48.5 identified in this Western population is higher than the traditional cutpoint of 40–45 reported in Asian surgical series [6, 12], possibly reflecting differences in body composition, comorbidity burden, and healthcare context. Cross-validation (mean 48.3) confirmed this threshold was not an artifact of data-driven selection, though external validation in clinical cohorts is warranted.

4.5. *Clinical Implications*

These simple, inexpensive indices can be calculated from routine laboratory data in virtually any clinical setting, aligning with ESPEN recommendations for routine nutritional screening in cancer patients from the time of diagnosis [1]. GNRI and CONUT, which incorporate BMI or total cholesterol, showed more stable, linear associations across the full follow-up period and may be preferable for long-term risk stratification when these additional measurements are available. A PNI threshold of 48.5 identifies a high-risk group with substantially shorter survival (6.6 vs 12.8 years median survival; RMST difference 1.97 years at 10 years), though this finding should be interpreted cautiously given PNI’s Bonferroni-non-significant primary result. The concentration of risk in the first 2 years after assessment suggests that early nutritional evaluation and intervention may have the greatest impact, consistent with the time-dependent attenuation observed for single-timepoint measurements. In the context of the GLIM framework, low values of nutritional indices may trigger formal malnutrition assessment, which has clearer intervention pathways. The significant PNI $\times$ physical activity interaction ($p=0.023$) in our exploratory analysis suggests that the association between better nutritional status and survival may be enhanced by physical activity, a modifiable factor warranting further investigation in prospective studies.

4.6. *Strengths and Limitations*

Strengths include population-based sampling with national representativeness, dual-cohort cross-validation across 30 years, and comprehensive temporal analysis (time-dependent Cox, landmark, RCS, RMST, competing risks). We also performed head-to-head comparison of biochemical, dietary, and inflammatory indices, E-value sensitivity analysis, and bootstrap C-statistic comparisons with absolute risk measures (RMST difference) that enhance clinical interpretability. Extended covariate adjustment and cohort-stratified analyses demonstrated robustness to socioeconomic confounding and cross-generational consistency.

Several important limitations must be acknowledged. First, cancer stage, grade, and treatment data—the strongest prognostic factors in GI cancer—are not available in NHANES public-use data. The observed PNI-mortality association may partly reflect confounding by disease severity, particularly given that NHANES does not provide reliable data on time since cancer diagnosis (prevalent cohort design). Sensitivity analyses excluding early deaths partially address reverse causation but cannot substitute for stage-adjusted analysis. Second, cancer diagnoses were self-reported without medical record or cancer registry verification, which may introduce misclassification, particularly for upper GI sites.

Third, the use of two NHANES generations with different albumin assay methodologies (bromocresol green in NHANES III vs. bromocresol purple in Continuous NHANES) may introduce systematic measurement differences, though cohort-stratified analyses showed consistent effects and inclusion of cohort as a fixed effect did not alter results. Fourth, nutritional indices were measured at a single baseline time point, preventing assessment of longitudinal changes in nutritional status, which may be more clinically relevant than a single snap-

477 shot. Fifth, the public-use mortality file provides only a 10-group cause-of-death
478 recode, limiting cause-specific analysis granularity.

479 Sixth, the sample composition (77% colorectal cancer) limits generalizability
480 to upper GI malignancies, and site-specific subgroup analyses were underpow-
481 ered (esophageal $n=15$, pancreatic $n=8$, gallbladder $n=6$). Seventh, the pooled
482 analysis of NHANES III and Continuous NHANES spans different treatment
483 eras; while formal interaction tests did not identify significant heterogeneity, sec-
484 ular changes in GI cancer management may introduce unmeasured confounding.
485 Eighth, adjustment for only three demographic covariates in primary models
486 leaves potential for residual confounding. However, extended models incorpo-
487 rating smoking status, a simplified Charlson comorbidity score, education, and
488 income-to-poverty ratio—applied to the subset with available smoking data—
489 did not attenuate the associations; PNI HR remained consistent (0.68, 95% CI
490 0.48–0.97, $p=0.035$), and CONUT and GNRI remained robust (all $p<0.001$).
491 These findings suggest that the PNI-mortality association is not primarily driven
492 by smoking or comorbidity confounding, though residual confounding by unmea-
493 sured factors (alcohol use, detailed frailty measures) cannot be excluded.

494 Ninth, the population attributable fraction estimate (13.4%) should be in-
495 terpreted with caution as it assumes a causal relationship between low PNI and
496 mortality, an assumption that cannot be verified in an observational design.
497 Tenth, the HEI-2015 analysis relied on a single 24-hour dietary recall, which
498 does not represent usual dietary intake and may bias associations toward the
499 null.

500 5. Conclusions

501 CONUT and GNRI are independently associated with all-cause mortality
502 in GI cancer patients after Bonferroni correction, with effects consistent across

two NHANES generations spanning 30 years. PNI showed a nominally significant association that did not survive multiple testing correction and should be interpreted as a secondary finding requiring independent replication. The protective associations are time-dependent (particularly for PNI, concentrated in the first 2 years) and operate primarily through non-cancer mortality pathways, suggesting these indices capture physiological reserve and resilience rather than directly reflecting cancer-specific biology. An optimal PNI threshold of 48.5 identifies a high-risk subgroup with a 1.97-year RMST difference at 10 years, though this requires validation. The prognostic value is driven predominantly by serum albumin, with lymphocyte count contributing minimal independent information, raising the question of whether albumin alone may suffice for risk stratification. Inflammatory markers (NLR, SII) show comparable performance, and joint assessment of PNI and NLR identifies a particularly high-risk group. These simple, inexpensive indices may facilitate early risk stratification and trigger formal nutritional assessment in GI cancer patients, though future studies with cancer stage, treatment data, and longitudinal nutritional measurements are needed to establish their independent prognostic value.

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Data Availability Statement

The data used in this study are publicly available from the National Health and Nutrition Examination Survey (NHANES) at <https://www.cdc.gov/nchs/nhanes/>. Analysis code is available from the corresponding author upon reasonable request.

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531 **Conflict of Interest**

532 The authors declare no conflicts of interest.

533 **Author Contributions**

534 **Zhuha Zhou:** Conceptualization, Methodology, Formal analysis, Investiga-
535 tion, Data curation, Writing—original draft, Writing—review & editing, Visu-
536 alization. **Yiqi Cai:** Supervision, Writing—review & editing. **Yongyu Bai:**
537 Investigation, Writing—review & editing. **Qigang Xu:** Data curation, Investi-
538 gation.

539 **Figure Legends**

- 540 1. **Figure 1:** Kaplan-Meier survival curves. (A) By PNI tertile (low, mid,
541 high); (B) By PNI threshold 48.5, with median survival and p-values.
- 542 2. **Figure 2:** Forest plots of adjusted HR per 1-SD. (A) All-cause mortality
543 (Cox); (B) Competing risks: cancer death vs non-cancer death (Fine-
544 Gray).
- 545 3. **Figure 3:** Time-dependent analysis. (A) Hazard ratios across follow-up
546 periods (0–2, 2–5, 5+ years); (B) Landmark analysis at 1, 3, and 5 years.
- 547 4. **Figure 4:** Restricted cubic spline dose-response curve for PNI (3 knots),
548 with 95% CI.
- 549 5. **Figure 5:** ΔC -statistic comparison. (A) PNI vs inflammatory markers
550 (NLR, SII); (B) PNI decomposition: albumin, PNI, and lymphocyte.

551 **6. Figure 6:** Subgroup forest plot—PNI effect across demographic and clin-
552 ical strata.

553 **Supplementary Materials**

554 **Tables S1–S7** and **Figures S1–S3** are available online as supplementary
555 data.

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